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Unit 2 - Bringing Ideas to Life

PDF Documents

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This PDF document is included in the unit materials.

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

BRIGHT LIGHTS, SAFE NIGHTS: JAMIE'S INTERSECTION INNOVATION

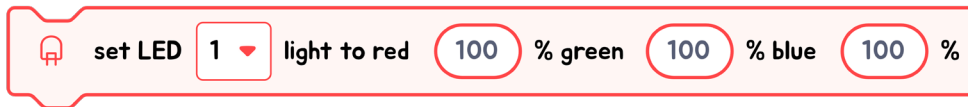
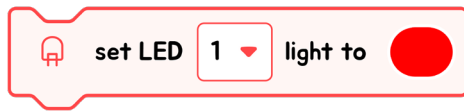
In the quiet suburbs where young Jamie lived, the community was growing concerned about a poorly lit intersection that had recently been the site of several minor accidents. Seeing this as both a challenge and an opportunity to test his engineering skills, Jamie decided to build a functional traffic light. He gathered LEDs, a small controller board, and a push button, setting up a temporary traffic signal to aid pedestrians and drivers until a permanent solution could be implemented.



Jamie worked on his project in his garage, connecting the red, yellow, and green LEDs to the controller board. He programmed the system so that the lights would change automatically but also included a manual override button for pedestrians needing to cross safely. His setup not only made the intersection safer but also caught the attention of local community leaders. Impressed by his initiative and technical prowess, they invited him to demonstrate his traffic light at the next community meeting, hoping to inspire other young inventors in the neighborhood.



ABOUT LED MODULE

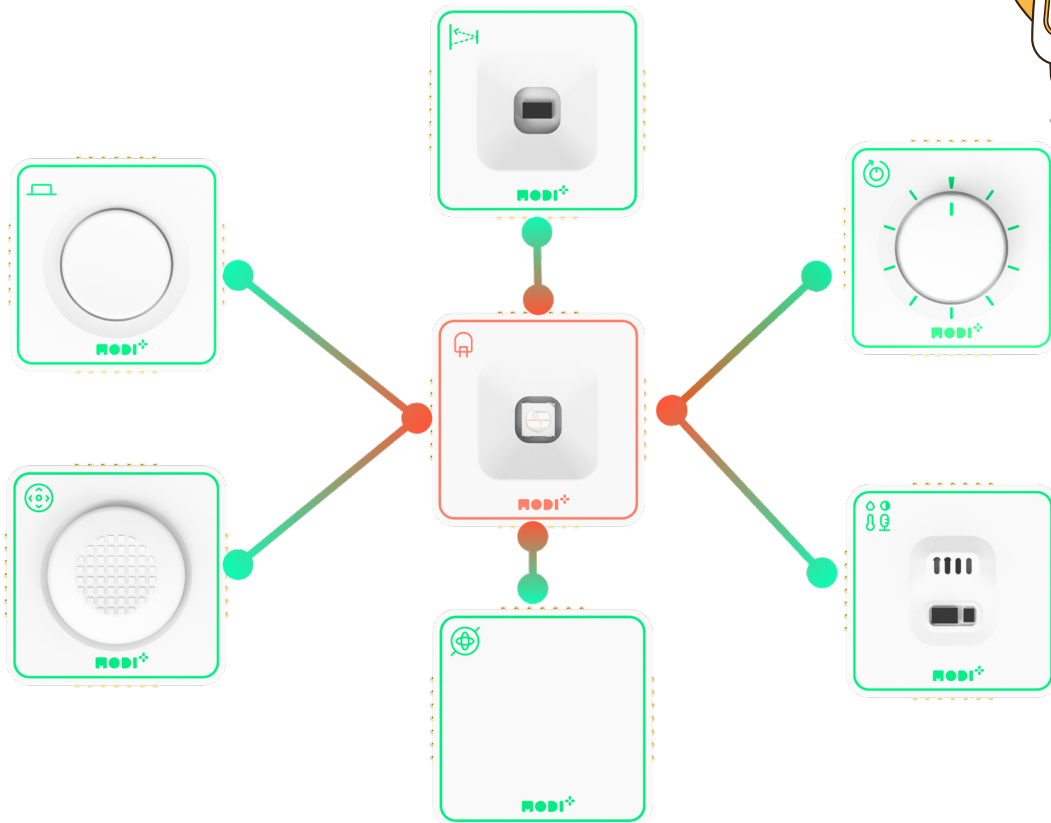


The LED module features three distinct blocks in the block area. The first block enables you to select a color using hue, saturation, and brightness values. The second block allows you to set the color based on RGB (Red, Green, Blue) values. The third block is used to turn off the LED module

INTERACTIONS WITH OTHER MODULES

As an output module, the LED can interact with all input modules. You can observe these interactions by connecting the LED module to any input module, even without any programming required!

INTERACTION BETWEEN LED AND OTHER MODULES





Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



Dial



Button



Environment



Joystick



ToF



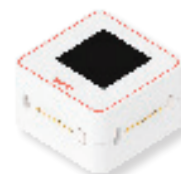
Speaker



IMU



LED



Display

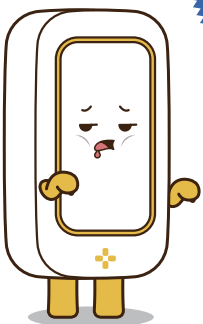
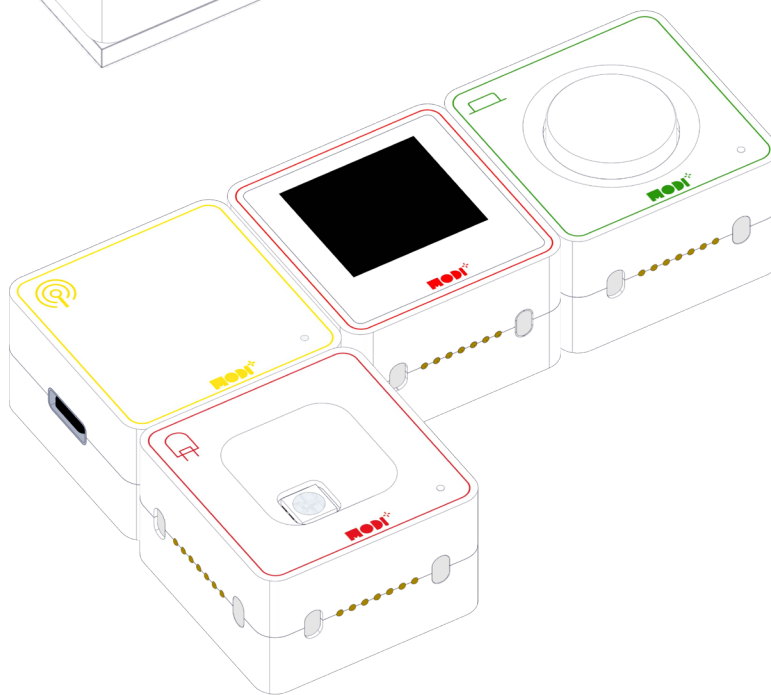
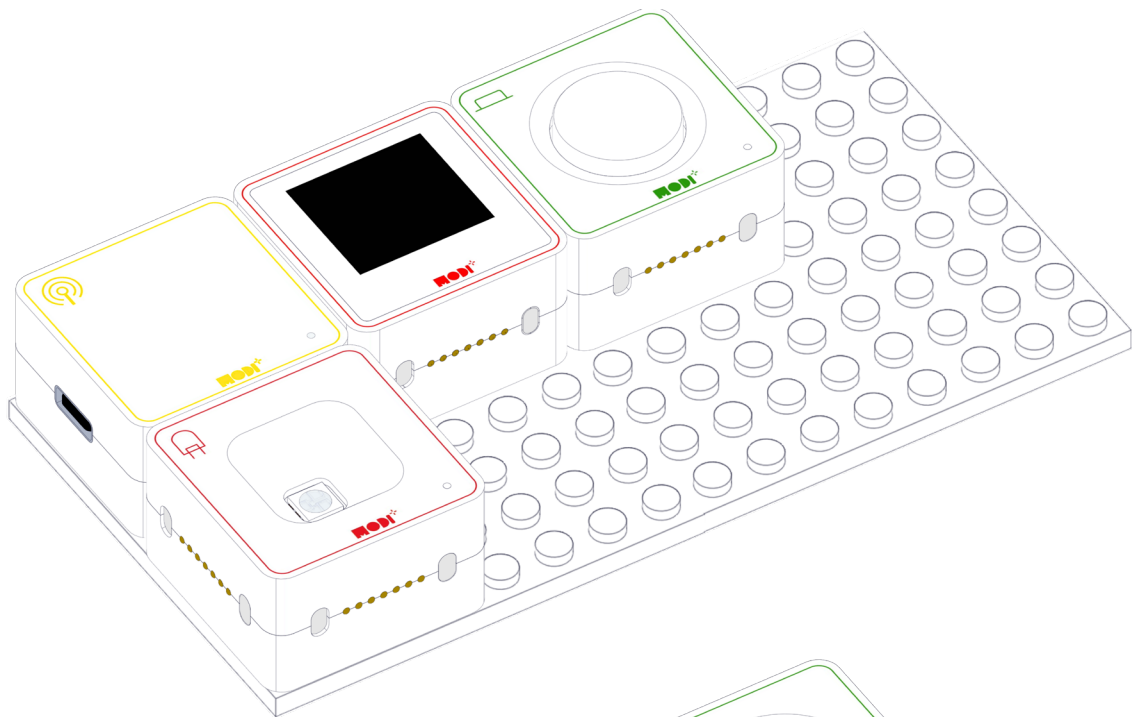


Motor A



Motor B

HOW DOES THE CREATION LOOK LIKE?

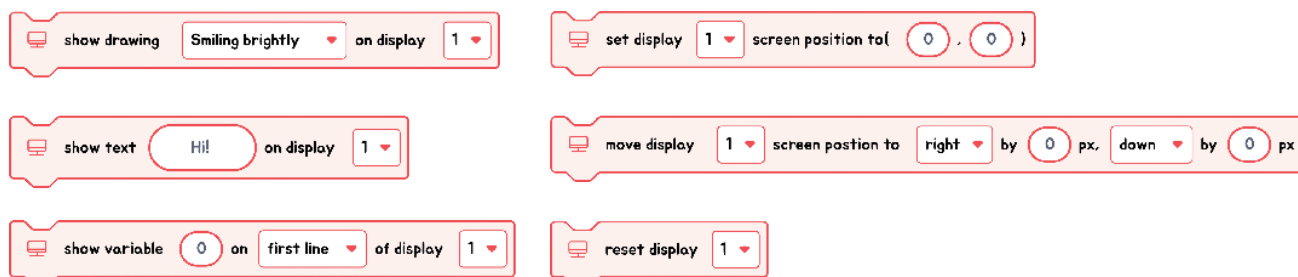
**TIPS**

The order of connections between the modules does not matter. As long as all the required modules are physically connected, they will function the same, regardless of which one is connected first.



Module Functionality

Let's check the details of the Display module below for this project.

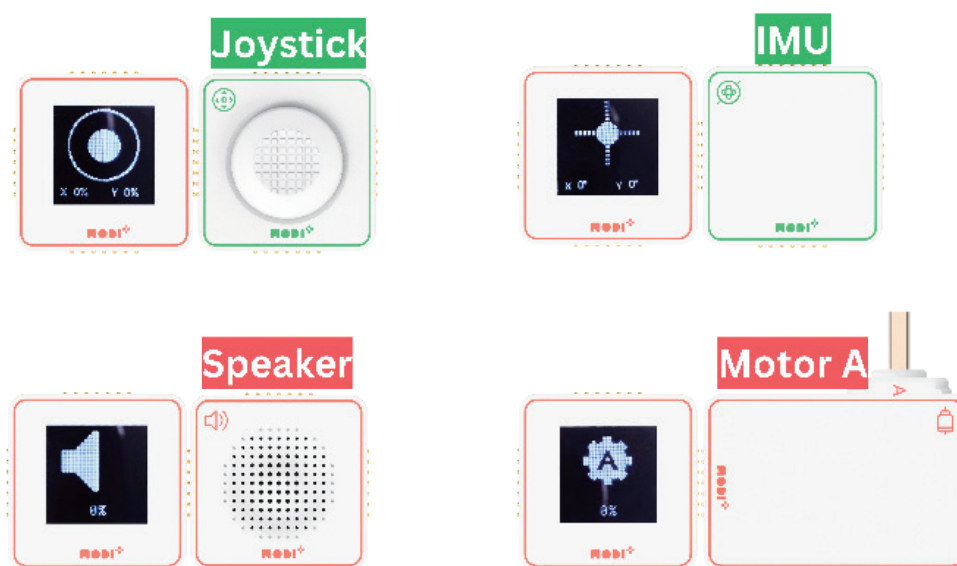


The Display module is like a tiny screen for your MODI projects. It uses 6 different blocks in the Code Editor to show pictures or information from other MODI modules nearby. So, when other modules talk to the Display, it can show you what they're saying or doing with images or data right on its screen!

AUTOMATIC DATA DISPLAY WITH MODI MODULES

The Display module is like the heart of a conversation among MODI modules. Imagine it as a screen that lights up with information whenever other modules, like the Button or the Light Sensor, have something to "say." When any input or output module is connected to it, the Display module automatically picks up their signals and shows what they're doing on its screen.

INTERACTION BETWEEN DISPLAY AND OTHER MODULES



So, if you press a Button module or cover a Light Sensor, the Display will show these actions without needing any special programming. It's like having a translator that instantly shows you what's happening in the MODI module language!